



Compliments of Hewlett-Packard

2nd Edition

HP Insight Control

FOR
DUMMIES®

Learn:

- About HP Insight Control and the new Gen8 Integrated Management features
- How you can use Insight Control to manage your servers and save time and money
- About the Insight Management product family

Bob Noller
Darrin Rawls
John Schmitz



About HP

HP creates new possibilities for technology to have a meaningful impact on people, businesses, governments, and society. The world's largest technology company, HP brings together a portfolio that spans printing, personal computing, software, services, and IT infrastructure at the convergence of the cloud and connectivity, creating seamless, secure, context-aware experiences for a connected world. More information about HP (NYSE: HPQ) is available at <http://www.hp.com>.

About HP Insight Control

HP Insight Control is essential server management that enables you to fully utilize the management features built into your HP ProLiant servers. Insight Control delivers powerful capabilities that allow you to proactively manage ProLiant server health (both physical and virtual resources), deploy ProLiant servers quickly, optimize power consumption, and control ProLiant servers from anywhere. Additionally, HP Insight Control extends the best managed ProLiant experience for customers who have standardized on HP IT Performance Suite, Microsoft System Center, and VMware vCenter.

***HP Insight
Control***
FOR
DUMMIES®
2ND EDITION

**by Bob Noller, Darrin Rawls,
and John Schmitz**



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Introduction



Today's data centers revolve around servers, lots of servers — both physical and virtual. And today's data center administrators are tasked with keeping these servers up and running, performing fast and efficiently. More than a few seconds of downtime (or even poor performance) can mean lost sales, incomplete data, and missed business opportunities. If you're in charge of your company's servers, you want them running at their fastest and most efficient.

This book details the challenges you face and what you can do to exceed the expectations placed on your data centers. Furthermore, you'll learn what HP Insight Control can do to make your job as easy as possible. It won't do everything for you, but it does allow you to automate as many processes as possible and get the information you need to keep your servers (and your supervisors) happy.

About This Book

This book introduces you to the concepts that surround modern data center management and the challenges those that manage those servers face. You'll understand the basic framework in which servers operate and what you need to do to make them work most efficiently.

Once you understand the problem, you'll move on to understand why HP Insight Control is the best integrated solution for managing the problem on HP ProLiant servers. This book takes you through all aspects of what's available in HP Insight Control and why you'll want to install it on your servers immediately (after finishing the book, of course).

This book contains six chapters that give you all the information you need to get acquainted with HP Insight Control. The first chapter outlines the key challenges facing data center administrators in general terms, the second chapter looks at efficiency and cost savings, and the rest of the chapters describe how HP Insight Control provides the tools you need to get the most out of every hour, watt, and dollar invested in your server infrastructure. This book was written with and for HP.

Icons Used in This Book

From time to time, you may see little icons off to the side of the text, resting comfortably in the margins. These icons aren't central to your understanding of the book, but they do call out some important information that enhances what you're reading about.



This icon gives you some helpful additional information. Think of it as the whipped cream and cherry on your informative ice cream.



A little reminder never hurts. This icon calls out information that reinforces a point made earlier in the chapter, and it's less awkward than a string around your finger.



You're obviously technical, or you wouldn't be reading this book to begin with. Still, this icon points out some information and settings that should appeal right to the geekiest aspects of your personality.

Part I

Infrastructure Management Basics

The 5th Wave

By Rich Tennant



"Well, I suppose we should plan on getting rid of those coal-burning servers."

In this part...

If you need to find out about the basics of infrastructure management, this is the part for you. We also cover some need-to-know info about HP Insight Control so you can get up to speed ASAP.

Chapter 1

Why Management?

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In This Chapter

- ▶ Understanding why you need management capabilities
 - ▶ Assessing the elements in your server infrastructure that require assistance
 - ▶ Maintaining an efficient and functional data center
-

You probably picked up this book because you already know the importance of device management in an IT environment. If you don't, just remember that time is money, and you want to save both. Getting an infrastructure device purchased, installed, configured, and running an application can be a daunting challenge for some people — and that's only the start of the real adventure. The lifecycle of the device can be the most complex, and expensive, activity you undertake — particularly if you don't have the right hardware and software solution working *for* you.

Are You Pained?

Have you ever been driving home, only to be called right back in to the office because an application went offline? Have you been on vacation and told to cut it short because an entire web application is down and you're the only one able to get that complex solution back up and running? Ever had the lovely experience of being woken at 3 a.m. by a phone call demanding you head to the office ASAP to troubleshoot a failure? If you have been around IT for any length of time, you probably have encountered these or other similar situations

that leave you thinking — why me? Surely there must be a better way to deal with this constant nag of failed hardware, software applications, and infrastructure components. Many times people look at software as an answer for some of these pains, evaluate it, buy it, and then watch it become shelfware when they don't have the time to actually implement and use it. Insight Control runs out-of-box with your HP ProLiant servers for an instant return on investment with a complete payback in just a few short months (while reaping benefits for years to come). It's also a fantastic tool for managing the rest of your infrastructure.



Insight Control comes in several flavors tailored to your IT environment. Maybe you want to integrate with an existing console in your IT space, like Microsoft System Center, VMware vCenter, Tivoli, CA Unicenter, or the HP IT Performance Suite. Maybe you want a complete infrastructure management system designed to be a single pane of glass for a better management experience. With Insight Control, these types of capabilities are enabled by the purchase of an Insight Control license for a server.

Servers and Storage and Networks – Oh My!

There are so many different technologies for a data center available today that are designed to solve the most complex business problems ever dreamed up on the planet. The IT environment becomes more complicated all the time with the explosion of virtualization, hypervisors, private and external cloud environments, and even completely external hosted IT environments. Still, deploying solutions like these doesn't have to be rocket science — you can handle them with standard solutions.

The core of a robust IT plan is the *right* tool for the job. That core isn't a *one-size-fits-all* environment that you have to deploy and manage. You must tailor an offering to an application delivery. Sometimes, that means a physical device; other

times a virtual device will work. Sometimes, you need a full-fledged storage array, and other times you only need some extended local storage. Regardless, you always need to know whether that system is up or down, serving up web pages or application sets, and whether the power draw or performance characteristics are within acceptable parameters for your users' experience. This is where management software like Insight Control comes into play, giving you a window into the core running pieces of the system along with access to control points to tweak or tune the environment as required.

Being productive in the data center depends on your needs:

- ✓ Are your systems running?
- ✓ Are you diagnosing and resolving issues quickly?
- ✓ Are you reactive or proactive?

Good management software gives you visibility into what is occurring at the server, proactively alerts you before systems fail, notifies you of the alerts, and then gives you the tools necessary to address the problem that may be next door or across the globe.



Don't forget the non-technical aspects of your system, either. Know the age of your system and when it leaves the comforting embrace of a warranty. Employees should track when a system leaves warranty and then facilitate a care pack or extended warranty session.

The Ultimate Connection?

As new systems roll out, you may be asked to handle a diverse server installation within a large data center, remote office, or sites with remote servers located elsewhere. You may also face your company buying other companies and having to integrate them from multiple locations. Remote access becomes key to keeping your IT sanity. If that call comes at the odd hour that an application is down, having the ability to get access to that server (regardless of where it is

located) is absolutely critical. Having the ability to remotely connect to a device over an intranet is a huge time and cost saver in comparison to physically travelling there.

The continued growth of equipment in the IT environment can pose problems in understanding what you have installed. Is it having any type of performance or failure condition? Is it covered under warranty? Are you getting abnormal performance statistics? Are you receiving hardware alerts? Any one of these items can lead you down a long road of different solutions (many of which are extremely complex!) trying to get visibility into your server farm. The good news is that HP provides an out of the box solution that addresses these kinds of problems.

Get Rolling!

Any type of application for a company requires the right server and storage environment built out to host the application set. This generally starts out as a server hardware purchase, and then turns into a deployment investigation. With many hardware vendors, you're left to figure out what drivers you need to install in a server to get it up and running with your particular operating system environment. This can quickly turn into a confusion of searching, reading manuals, piecing together scripts, praying, and general frustration. Not only do you need the server to run an operating system — you need to have some type of storage configured in order to either install the operating system or have active logical unit numbers (LUNs) for data. Many times this means that you're digging around looking for various device drivers. Insight Control contains preset deployment jobs, to deploy operating systems to all types of HP servers — no changes needed. Marry that up with the ability to boot locally or boot-from-SAN, and now you have a powerful deployment engine at your fingertips.

Do You Have the Energy?

As the IT industry has matured (over the last decade especially), the energy consumption of various components has grown and (in some cases) become a lot more efficient. Some products like the HP ProLiant server line have had a design goal of being extremely efficient in the energy consumption within the server itself, creating a large gap in actual power draw of a server as compared to a rated power draw on a power supply. IT environments that base their power and cooling capabilities off rated wattage for power supplies severely overprovision power in a data center and have a lot of very expensive trapped resources. Many data centers may not worry about power initially, but need more than 1.5 times the cooling necessary to handle 1 watt of energy. That's when the rubber meets the road — cooling capability is the first thing that runs out. Then you have to deal with an all-hands-on-deck effort to figure out how to reduce and consolidate resources to regain capacity. Face it, data centers are very expensive, and building a new facility is a huge drain on resources. It's always better to improve your current facilities than to add more.



You can use Insight Control to understand *exactly* what your power draw is for a server or rack run of servers. You can isolate hotspots in the rack, row, or even the entire data center. You see how power and temperature aspects have trended over the last 24 hours, and you can even store and track this information for up to three years.

Using Insight Control

For example, you can have a Chiller fail — now what? You want to shut down all your test and development systems and push systems running your website down to the lowest power state to keep the website running (albeit at a reduced performance). You need the database to be running,

so the power is only stepped down for those systems to reduce the power draw but keep them at an acceptable level. Insight Control gives you the ability to define the rules to implement one-step control to keep your critical systems running should this occur in your data center.

Where's the Bottleneck?

Many people look at a performance analysis as taking performance statistics from the operating system, manipulating the data, and then analyzing the environment for application bottlenecks. Although this is a necessary aspect of application management, it isn't the only performance analysis that needs to be accomplished. Servers running the operating systems need to be analyzed for performance bottlenecks and monitored closely for performance issues within a hardware component. Hardware performance analysis allows for analysis of various system subcomponents (such as a network interface card, processor, memory, or storage) and then it allows alerting and notification of an issue.

Operators need this capability to identify the problem when a user or customer reports a slow or failing application. That operator needs to know whether the issue comes from the hardware level or at the application layer. If everything is okay from a hardware perspective, you can move on and focus your triage efforts at the application or operating system level.

Time-Tested Approaches

Managing devices in an IT environment dates back decades. Mainstream management applications like Simple Network Management Protocol (SNMP) have been the mainstay of management protocols for a long time and are still evolving. Still, no control functions (such as SNMP Writes) should be done over the network for security reasons.



When managing servers, you must address security issues. Control operations should be performed via a secure mechanism like secure shell (SSH) or secure sockets layer (SSL) communications. Security is a key concern in any company, but with various government or industry regulations, security aspects are critical. The SNMP protocols might be used for read-only activities, but other protocols like WBEM, SSH, and SMI-S (WBEM for storage) might be used to actively control infrastructure devices. Whatever protocols are used, security issues must be addressed in the management space.

Chapter 2

Get More with HP Insight Management

In This Chapter

- ▶ Learning the benefits of converged infrastructure and unified management
 - ▶ Examining what HP has to offer
 - ▶ Saving money and power with a well-configured system
-

It's not enough to provide sufficient storage for your server needs. System administrators need to optimize their hardware and software operations as much as possible in order to produce the best possible data services while saving as much money as possible. This chapter details how HP's solutions make server management efficient and cost-effective while giving you the best possible servers.

Understanding Converged Infrastructure

If your company is like most, you've added servers, storage, and networking devices to keep pace with your business demand. As much as 70 percent of IT budgets are spent on operations versus innovation.

You manage everything separately with a network stack, a server stack, virtual machines, a storage stack, a management stack — all built up and siloed in their own worlds. No one really talks to each other (like your sometimes-awkward break room).

To compete and grow in a market, companies must be able to move rapidly to seize new opportunities, reach new customers, and grow market share. This is what the HP Converged Infrastructure is about.

Simply put, a converged infrastructure overcomes the inflexibility and high costs created by data center, application, and IT sprawl (and no one talking to each other). You can shift/invest more of your resources (such as people and money) into innovation. With an HP Converged Infrastructure, compute, storage, network, and other resources are integrated by design for *unified management* using HP Insight Management (see Figure 2-1).

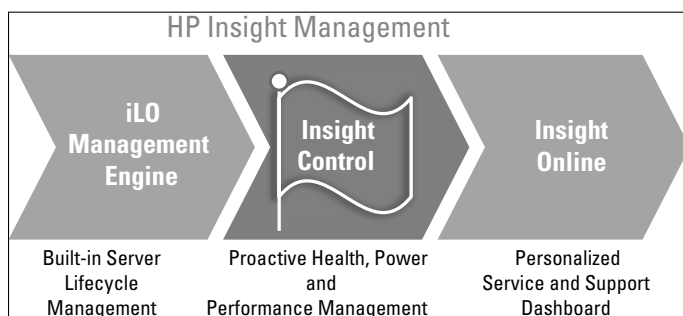


Figure 2-1: HP Insight Management.

Looking at the Integrated Tool Set

Face it — management is a chore if you have different management tools that show you only one part of your infrastructure. Not only is it inefficient and costly, but multiple, disparate tools tend to give you only data. You really need *information*, which you can only get with an integrated management solution.

Unified management requires integrated tools that work together to manage your IT infrastructure. The HP Insight Management portfolio helps you understand, manage, and optimize your HP ProLiant, HP BladeSystem, and HP Integrity Server resources in real time over a long period.

iLO Management Engine

This set of embedded management comes with your HP ProLiant servers and supports the complete lifecycle of the server from initial deployment through ongoing management to service alerting.



Find out more at www.hp.com/go/ilo.

Insight Control

Insight Control is essential management that saves you time and money by enabling you to proactively manage ProLiant health and performance, deploy ProLiant servers quickly, optimize power confidently, and completely control your ProLiant infrastructure from anywhere. Additionally, Insight Control integrates seamlessly into Microsoft System Center and VMware vCenter Server consoles to extend the best ProLiant management experience in those environments.



Find out more at www.hp.com/go/insightcontrol.

Insight Online

Insight Online makes it easier to find the information you need when you need it. Insight Online provides secure, personalized, one-stop access to product and HP support information, such as contracts and warranties specific to your IT environment. With Insight Remote Support integration, devices and configurations can be displayed, and service events and support cases can be tracked from anywhere at anytime.



Find out more at www.hp.com/go/insightonline/info.

Examining Unified Management

The unified management integrated in the HP Insight Management portfolio gives you the ability to:

- ✓ Make more informed decisions — not just best guesses.
- ✓ Accomplish more in less time — be home for dinner.
- ✓ Deliver better service to your business — get kudos instead of complaints.

Having HP's unified management allows you to address the server management challenges you face today.

Your company already spends a lot of money on IT, and needs to keep pace with your growing business. In fact, server management costs eight times as much as buying servers does. In general, key factors include:

- ✓ Time associated with server deployment and migration (getting servers up and running)
- ✓ Server downtime and the cost associated with lost user productivity
- ✓ Energy costs
- ✓ Remote server management



HP Insight Management, specifically HP Insight Control, is designed to help address these customer issues and reduce the amount you spend on server management.

Part II

Insight Control Answers

The 5th Wave

By Rich Tennant



"They can predict earthquakes and seizures,
why not server failures?"

In this part...

If you have questions about HP Insight Control, you'll find the answers in this part of the book.

Chapter 3

Deeper into Insight Control

In This Chapter

- ▶ Deploying and migrating
 - ▶ Monitoring health
 - ▶ Managing performance
 - ▶ Supporting servers
 - ▶ Optimizing power
 - ▶ Controlling servers
-

You can implement server management in a wide variety of ways — using ad hoc methods with homegrown software, off-the-shelf integrated software, or fully customized integrated solutions. HP Insight Control offers an integrated software solution that helps administrators excel in their server management tasks.

HP Insight Control provides essential infrastructure management and enables the management capabilities built in to HP ProLiant servers. It provides a single, integrated management solution for both physical and virtual servers, and supports agentless management on HP ProLiant Gen8 servers.

Insight Control uses a central management console, providing a comprehensive and intuitive interface for running an entire set of server management tasks. And it also offers the flexibility to integrate seamlessly into Microsoft System Center and VMware vCenter Server environments if the user has already standardized on those management platforms.

Deploy and Migrate Quickly

HP Insight Control makes it easy for you to deploy new ProLiant servers in your business, and to migrate applications and data from your current servers to your new servers as quickly as possible.

- ✓ **Server deployment** capabilities automate the process of deploying and provisioning server software.
- ✓ **Server migration** provides an unlimited number of migrations — get your existing capabilities up and running on new servers or implement virtualization.

Features

Insight Control supports deploying servers either through imaging or through scripting. It also supports deploying Windows, Linux, VMware ESX, and Microsoft Hyper-V systems.

Server deployment

Insight Control server deployment is an easy-to-use, drag-and-drop solution for deploying standard server configurations and software builds from a remote console. It runs in the Windows, Linux, and Microsoft System Center operating systems out-of-the-box. You can automate hardware configuration through operating system and application deployment.

Imagine creating a template to deploy to a new virtual machine in your network. Then assign a single task within Insight Control to automatically update firmware and operating system components for physical and virtual servers in your network.



Software optimizations for HP servers are also provided in the form of integration modules, which include scripting toolkits, configuration jobs, sample unattended files, software drivers, management agents, and documentation. You can also use deployment events and scripts to automate the process of server configuration and software deployment.

Server migration

Insight Control server migration (see Figure 3-1) enables you to replicate your current server to a new physical or virtual server in an automated, affordable way. Migrations are wizard-driven, which reduces the potential for any errors. You can perform unlimited migrations of various types, including:

- ✓ Physical-to-ProLiant (P2P)
- ✓ Virtual-to-ProLiant (V2P)
- ✓ Physical-to-Virtual (P2V)
- ✓ Virtual-to-Virtual (V2V)

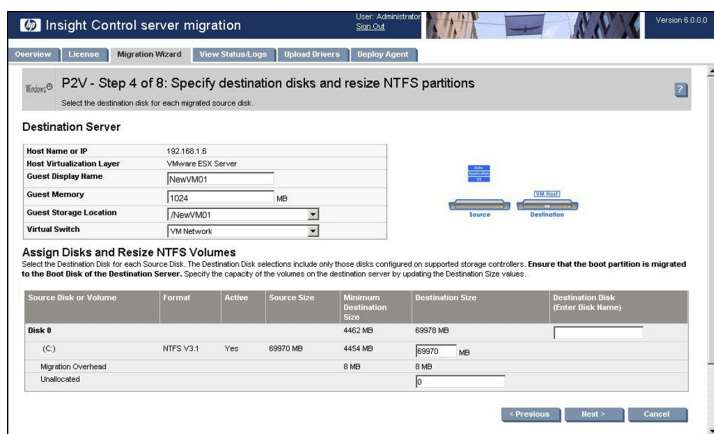


Figure 3-1: HP Insight Control server migration.

Migrate from any x86 or x64 server running a supported operating system or a supported VM-type. The destination can be any supported HP ProLiant or BladeSystem server or a supported virtual machine — or more than one!



You can migrate between virtualized environments like VMware ESX and Microsoft Hyper-V virtual machine environments. Insight Control automates the transfer of operating systems, applications, and data from your old server (or VM) to the new ProLiant server (or VM) without errors.

Insight Control supports SAN Boot or SAN Attach deployment on many ProLiant servers, and server migrations across iSCSI

and FibreChannel-over-Ethernet (FCoE) environments and Linux SAN environments.

Benefits

Deployment and migration features in Insight Control can be good for your entire environment:

- ✓ Simplify automated server deployment, allowing rapid response to changing workloads
- ✓ Improve overall consistency of configurations across multiple servers for greater accuracy
- ✓ Eliminate the need for IT personnel to be physically present by establishing virtual presence
- ✓ Maximize productivity and decrease Total Cost of Ownership by automating server configurations and software deployment
- ✓ Save time and money by performing unlimited migrations from old servers to new servers

Monitor Health Continuously

HP Insight Control is integrated with HP Systems Insight Manager (HP SIM). It is a unified and secure platform to centrally manage servers, storage, and other infrastructure devices across many operating system environments. HP SIM is the basis for health management for your IT environment.

When integrated with HP SIM, Insight Control can monitor your entire infrastructure, view both physical and virtual servers from one location, and be alerted to any potential failures or failures that are in progress. HP SIM is built to industry standards, supports agentless management of ProLiant Gen8 servers, and can be extended using custom applications and scripts to meet your needs (see Figure 3-2).

Features

Insight Control, using HP SIM, provides foundational management services for the hardware-level and supports multiple operating systems on servers and storage. It offers the following benefits:

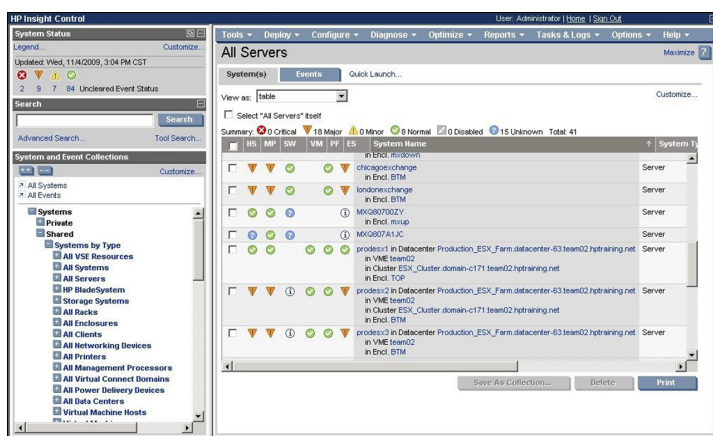


Figure 3-2: HP Systems Insight Manager (dashboard).

- A single management view of the infrastructure across different operating systems
- Management of all HP server and storage platforms
- Fault, configuration, and asset management
- Version control, with software and firmware baselines, integrated with the HP Smart Update tool
- Role-based security
- Automatic discovery, inventory management, reporting
- Monitoring and management of health and alerting
- Contract and warranty information
- Access to Remote Support and Support Automation
- Integration with other solutions and customized tools

HP SIM allows system administrators to search across several different management consoles, with role-based administration preserved across all systems (see Figure 3-3).



HP SIM can be used to manage smaller environments of one to several hundred devices or larger environments of server and storage farms with several thousand devices.

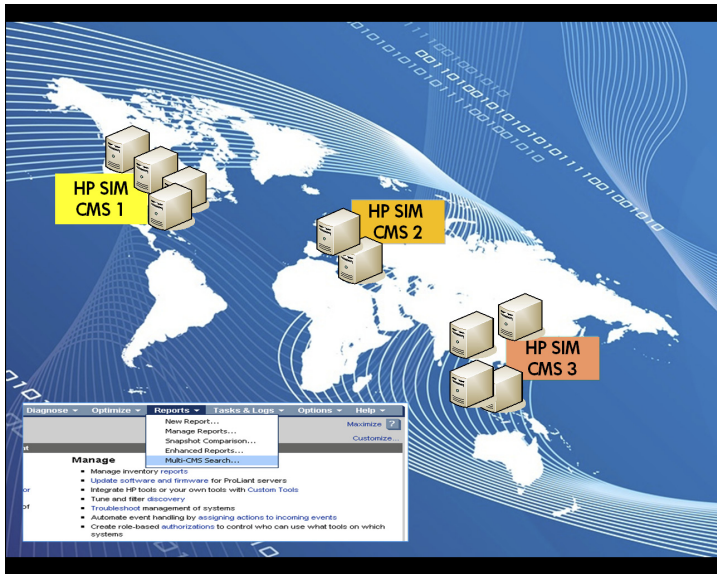


Figure 3-3: HP Systems Insight Manager (search across consoles).

Users with Windows-based servers can manage servers running a mixture of operating systems including: Windows, Linux, HP-UX, OpenVMS, and others.

Benefits

HP Insight Control, based on HP SIM, provides you with health-monitoring tools for your immediate business benefit:

- ✓ An easy, simple, inexpensive way for system administrators to maximize system health and uptime.
- ✓ Get your infrastructure up and running quickly.
- ✓ Avoid downtime by proactively monitoring for problems, analyzing and identifying them quickly, and issuing alerts to take corrective actions promptly.
- ✓ Move virtual machines before your server hardware fails in either the VMware or Microsoft Hyper-V environments.
- ✓ Streamline your processes and reduce the complexity of your operating system-server-storage environment.

Manage Performance Proactively

HP Insight Control detects, analyzes, and manages hardware performance problems on HP ProLiant servers. It provides the tools you need to receive proactive notification of potential bottleneck conditions and to debug performance issues.

Features

HP Insight Control addresses performance bottlenecks on hardware devices like processors, memory, network adapters, disk subsystems, and PCI buses. It also monitors hardware performance continuously — comparing actual performance with the known capabilities of each component, and determining (in real time) which components are experiencing performance issues. It can detect and display a warning before an actual bottleneck occurs, as shown in Figure 3-4. It can log server performance data and replay it in offline analysis, report on server performance utilization, and analyze and report on component configurations to determine problems.

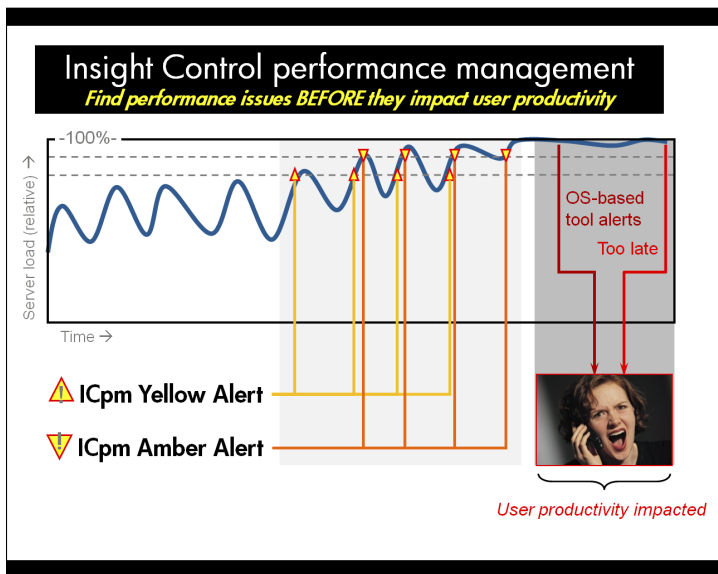


Figure 3-4: HP Insight Control performance management.

Benefits

HP Insight Control provides you with performance management tools for your immediate business benefit:

- ✓ Avoid downtime by taking corrective actions promptly.
- ✓ Be more productive by pinpointing and resolving performance bottlenecks before they become problems.

Monitor Servers Remotely

HP Insight Remote Support automatically provides secure, 24x7 remote support for your HP servers and storage. This software leverages standard Internet security protocols, and principles so you can monitor your systems remotely for hardware failure. In many cases, you can avoid problems before they occur.

HP Insight Online utilizes HP Insight Remote Support to provide you with secure access through an online personalized dashboard for monitoring your infrastructure from anywhere, at any time. Insight Online allows you to monitor device health, service events, and support status; and to manage your HP support contracts and warranties online.

Features

HP Insight Remote Support and HP Insight Online offer you choices for better management and support.

HP Insight Remote Support

HP Insight Remote Support (HP Insight RS) offers comprehensive remote support on HP servers, server blades, storage blades, and storage.

The console integrates with the HP SIM console and allows an IT operator to set boundaries and thresholds of the remote IT environment.

HP Insight RS provides many essential capabilities:

- ✓ Single Management Console with serviceability attributes
- ✓ Detailed fault diagnostics/analysis of the hardware environment
- ✓ 24x7 remote monitoring of your hardware environment regardless of your IT staff coverage hours
- ✓ Automatic event detection, support case generation, reporting, and routing without manual intervention
- ✓ Alerts to accurately diagnose and receive support
- ✓ Pre-failure advisories
- ✓ Warranty and Service Contract entitlement management
- ✓ Service status and detailed problem resolutions
- ✓ Proactive capabilities for HP Proactive Select users

HP Insight Online

HP Insight Online is part of the HP Support Center and is seamlessly integrated with Insight Remote Support 7. It provides you with quick, easy, and secure access from anywhere to relevant product and service information to save you time and hassle.

HP Insight Online provides essential capabilities:

- ✓ Personalized dashboard for monitoring device health, service event, and contract and warranty status
- ✓ Visual display of devices, device groups, and configurations
- ✓ Online monitoring of hardware service events and related support cases
- ✓ Proactive contract and warranty tracking
- ✓ Device information can also be shared with HP Authorized Partners



Remote support requires access to an encrypted outbound-initiated Internet connection (HTTPS/443), which is secure enough for most company security policies.

HP Insight RS supports SNMP/WBEM monitored devices and integrates into a centralized, dedicated HP SIM server, and supports installation on a VMware guest.

Benefits

HP Insight Online with Insight Remote Support provides:

- ✓ Remote, real-time, phone-home-to-HP 24x7 support
- ✓ Reduced intervention and increased control
- ✓ Accurate resolution in less time
- ✓ Personalized access: anywhere and any time



HP Insight Remote Support comes at no extra cost as part of your warranty, HP Care Pack Service, or contractual support agreement with HP. And HP Insight Online is freely available to HP ProLiant users.

Optimize Power Effectively

High costs associated with power and cooling are key issues for many IT environments today. Common energy problems that customers encounter are: downtime, running out of power or cooling capacity, and hot spots.

HP Insight Control provides centralized monitoring and control of power consumption and thermal output. It uses real-time information about power and cooling usage in your environment to allow you to manage energy. It also increases the capacity of your data center by reducing power and cooling requirements to accurately reflect the workloads of your servers. Look at Figure 3-5 for an example.

Features

HP Insight Control provides a central location where power and thermal data for tens, hundreds, or thousands of servers may be viewed and managed. You can control the resources of your IT environment in different ways:

➤ **Power Measurement** provides real-time, around-the-clock power consumption data for superior control over operational costs. It can display three years of power consumption and server temperature data and it can measure power consumption for heterogeneous environments. See Figure 3-6.

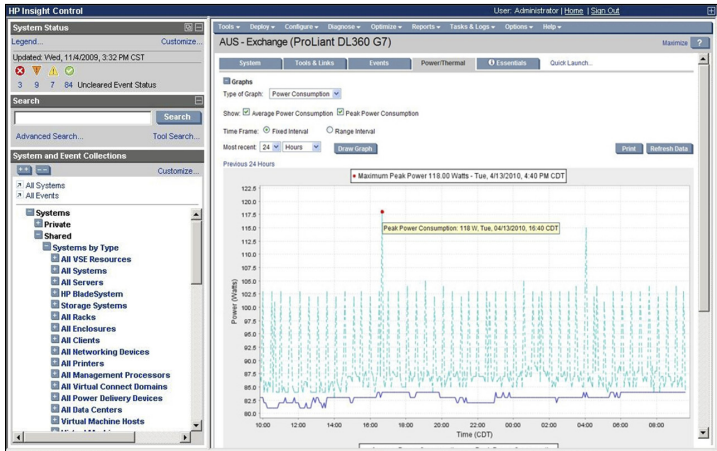


Figure 3-5: HP Insight Control power management (power measurement).

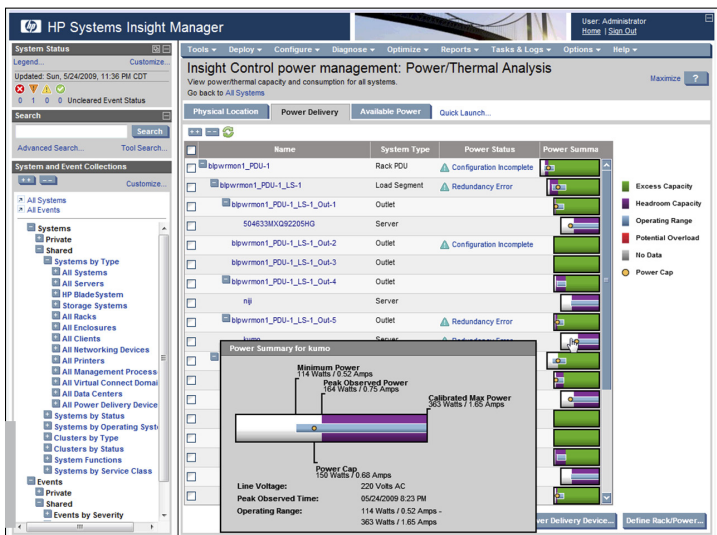


Figure 3-6: HP Insight Control (power and thermal analysis).

- ✓ **Power Regulation** conserves power without performance impact on a single server or multiple servers, using technology that's built into the server. Action scheduling can also set time-based power policies.
- ✓ **Dynamic Power Capping** leverages the power measurement history to safely limit power usage using a fast-acting, hardware-based algorithm — without performance degradation and without risk to your electrical infrastructure. You can also use that data to understand the cost of powering and cooling servers, and export data to other reporting tools.
- ✓ **Intelligent Power Discovery** is an energy-aware network between IT and facilities. It allows you to safely fit more servers into the existing power and cooling budget by accurate and error-free planning. You can monitor and manage both HP and third-party servers to automatically discover and map new servers in your environment. It detects configuration errors, common wiring problems (redundancy errors, potential overloads, and unauthorized equipment), and servers and services that are at-risk due to hotspots and electrical and thermal overloads.
- ✓ **Facilities Integration** enables IT managers to communicate with facilities managers, using measured data instead of estimates, to resolve power and cooling problems and to set realistic plans. Partner products for facilities planning, such as nlyte Software and Eaton Forseer, use this data for better control.
- ✓ **Data Center Power Control** reduces energy costs and takes control of power and cooling during catastrophic events. Your predefined rules and scripts can reduce power consumption during critical times, reduce the power to noncritical servers, and keep critical servers running (if possible). It can also save money on energy usage during times of peak demand.
- ✓ **Reports** (Crystal format) for power and thermal data can be scheduled and e-mailed on a weekly (or other periodic) basis. A specific *underutilized servers* report identifies problem servers for further action.

Benefits

Insight Control can reduce energy usage, efficiently use existing power/cooling capacity, and extend the life of your IT environment. It can:

- ✓ Accurately identify and eliminate areas of power waste.
- ✓ Increase uptime and eliminate human error by using automatic power discovery during server installations.
- ✓ Improve energy allocations.
- ✓ Safely fit more equipment in your existing environment's power and thermal capacity without affecting performance or increasing the electrical risk.

Control Servers from Anywhere

HP Insight Control, featuring iLO Advanced software, provides comprehensive lights-out remote management for ProLiant servers. HP iLO Advanced software enables the best capabilities of the HP iLO Management Engine. HP ProLiant servers equipped with the HP Integrated Lights-Out (iLO) management processors allow you to take control of your servers from anywhere.

Features

HP provides standard iLO functions for HP servers without additional software. Basic system board management functions, diagnostics, and standard power controls are functional regardless of the servers' state of operation. These virtualize system controls to help simplify server setup, provide health monitoring and power and thermal control, and enable remote server management, as seen in Figure 3-7.

Distributed teams can also collaborate to solve complex IT problems without being physically present on the site.



Insight Control allows you to deploy and update servers remotely by mounting DVD, CD, and USB media (or ISO images and folders) on your client system or management station. Actions on remote servers by scripts can specify dates, times, servers, and so on.

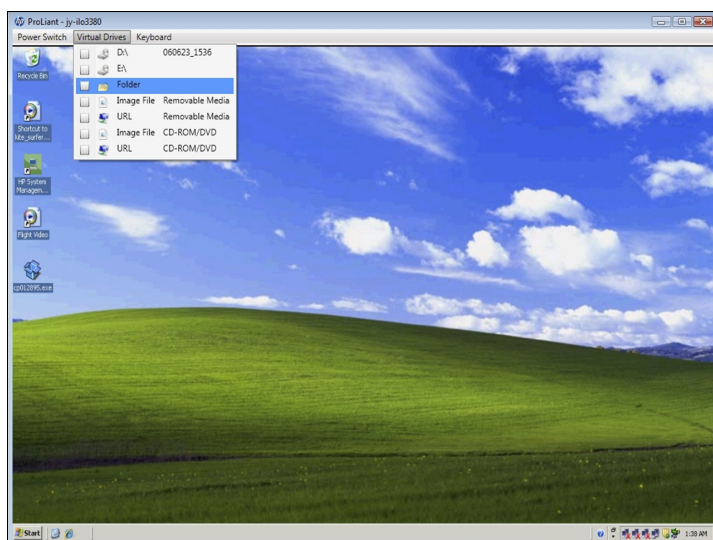


Figure 3-7: HP Insight Control remote management (remote server screen).

Insight Control can also integrate with enterprise-class directory services, like Microsoft Active Directory and Novell eDirectory, to authorize directory users with assigned roles to processors.

Benefits

HP Insight Control remote management, using iLO Advanced software, allows control of servers from anywhere:

- ✓ Troubleshoot and collaborate from any location whether the operating system is running or offline.
- ✓ Spend less time and money on travel by administering remote servers without having to visit them.
- ✓ Collaborate, diagnose, and fix problems quickly by sharing remote console sessions with up to six administrators and by playing back captured video footage.
- ✓ Speed up collaboration and system recovery times by using remote consoles and virtual media capabilities.

Chapter 4

Integrating Insight Control with Enterprise Management Products

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In This Chapter

- ▶ Working with Microsoft System Center
 - ▶ Working with VMware vCenter Server
 - ▶ Working with HP Operations Center Manager
-

HP designed Insight Control to work with leading enterprise management tools, including Microsoft System Center, VMware vCenter Server, and HP IT Performance Suite. You no longer have to open different management consoles to switch between HP server management and enterprise management.

Microsoft System Center and VMware vCenter Server integrations are included with HP Insight Control. If you're new to Insight Control, you can purchase it as a single license or in bulk quantities. If you buy ProLiant and BladeSystem servers, Insight Control may come bundled with that equipment. And if you're an existing Insight Control customer and you're under a current Software Updates contract or if you're managing HP Storage, you can download the integrations free of charge.

Microsoft System Center

Insight Control extends Microsoft System Center with the following tools and services, and each one benefits you and your business — and now with support for System Center 2012.

System Center Operations Manager (SCOM)

Insight Control extends SCOM to monitor the health and configuration of your HP servers running Windows, Linux, VMware, and BladeSystem enclosures, as well as ProLiant Gen8 agentless servers. You can also access iLO, the System Management Homepage, and the BladeSystem Onboard Administrator directly from SCOM. Figure 4-1 shows an alert notifying you that an error has occurred.

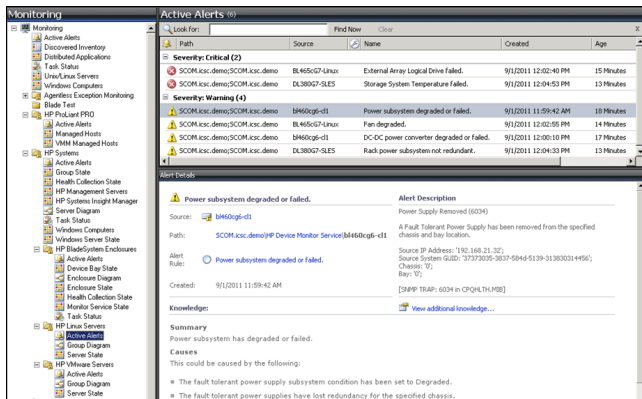


Figure 4-1: Health management in SCOM.

System Center Configuration Manager (SCCM)

Insight Control extends SCCM OS Deployment (OSD) to deploy Windows Server to bare-metal HP servers, including pre-deployment hardware and BIOS configuration and post-OS driver and agent installation. This allows an unattended installation from within SCCM, so you don't have to prep the hardware first and then install the operating system manually.

Insight Control also extends SCCM to deploy HP server firmware, drivers, and management agent components throughout the network using the same functionality as OS and application patch deployment. Deployment of any HP components goes much more quickly and reliably.

Performance Resource Optimization (PRO)

Insight Control and PRO respond automatically to events on virtualized server hosts. System Center Manager Virtual Machine Manager (SCVMM) can respond to pre-failure conditions intelligently by showing a PRO tip (shown in Figure 4-2) that provides automated virtual machine migration.



Quick action like this prevents potential hardware issues from causing downtime!

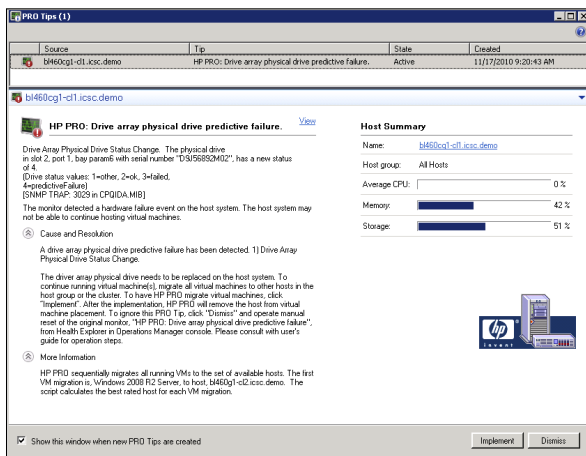


Figure 4-2: A sample PRO tip in SCVMM.

VMware vCenter Server

HP Insight Control for VMware vCenter Server is a single management plug-in from HP that allows you to monitor and manage HP servers, networks, and storage from within the vCenter management console. It appears as a tab on the main screen of the management console. The tab shows context-sensitive server, networking, and storage-specific information for the individual elements of your VMware virtualized infrastructure (see Figure 4-3).

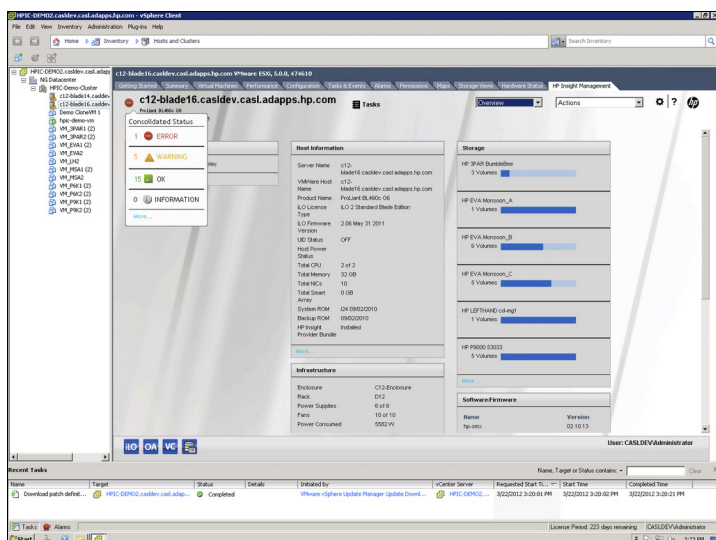


Figure 4-3: HP Insight Control for VMware vCenter server integration.

How do these features result in greater productivity? Your VMware administrators can monitor the health, power, and performance of virtual machines and their host systems. Your team can monitor operations at the cluster, server, or resource level from a single pane of glass.



You can monitor and manage operations from work, home, or vacation using secure wireless.

Here are some benefits:

- View hardware events from within the vCenter console and set up alarms to respond to events automatically. And your VMware administrators don't have to learn a new user interface.
- Visually trace and monitor your network from the host to the individual network modules for easy review and changes, as shown in Figure 4-4.
- Integration with the vCenter console allows you to manage and troubleshoot HP servers remotely by launching iLO, HP Onboard Administrator, or SIM.

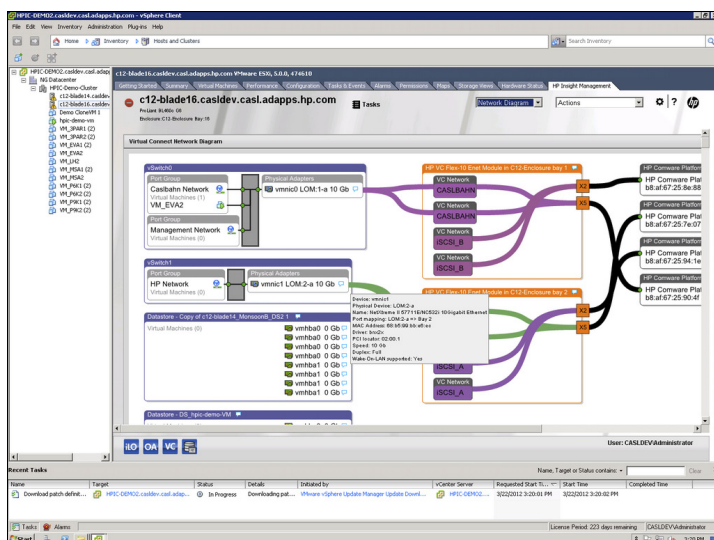


Figure 4-4: An end-to-end networking diagram.

- ✓ When new capacity is needed, you can deploy new hosts from bare metal and continuously control stability by managing firmware updates online.
- ✓ You can provision, configure, and monitor your HP Storage arrays from your vCenter console, which allows you to create or clone virtual machines and create or expand your data storage, combined with ongoing health monitoring, as shown in Figure 4-5.

HP Operations Manager (HPOM) and HP SIM

HPOM software consolidates and correlates fault and performance events across your entire physical and virtual infrastructure. It enables you to create more stable business operations, and synchronize IT resources and business needs.

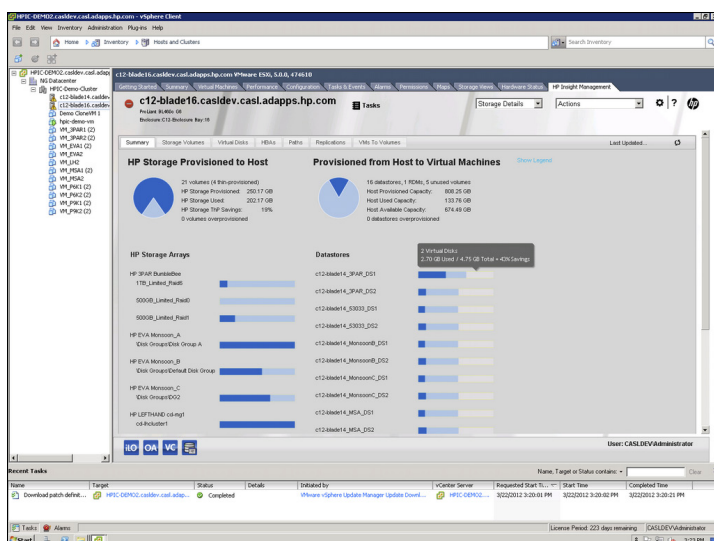


Figure 4-5: HP Storage management.

The Insight Integration for HP Operations Manager provides you with the option of receiving events directly from the managed device or from HP Systems Insight Manager.

You can also filter events yourself through HP SIM.

And if you're using HP SIM with Insight Remote Support Advanced, another option allows you to forward only the incidents that need physical intervention.

UCMDB and HP SIM

HP Universal Configuration Management Database (UCMDB) software stores, controls, and manages software and infrastructure components — and their associated relationships and dependencies. This improves and simplifies change control to avoid service disruptions. HP UCMDB software reduces mean time to repair to keep operations running more efficiently.

HP SIM integration enables you to automatically populate UCMDB with all the SIM asset data and simplifies configuration and SIM synchronization between HP SIM and UCMDB.

Chapter 5

HP Insight Control Licensing and Packaging Basics

.....

In This Chapter

- ▶ Learning the basics about software licensing
 - ▶ Purchasing your software license
 - ▶ Getting any support and help you might need
-

You will find HP Insight Control easy to purchase and deploy on HP servers. But how do you get started? How is Insight Control licensed and packaged? This chapter navigates licensing and packaging basics and tells you how to get software support and software updates for HP Insight Control.

Software Licensing Basics

In general, a software license gives you the legal right to use software according to specific terms, conditions, and restrictions. And HP Insight Control is no different, requiring a software license whether Insight Control is purchased with a new HP server or separately.

You will find the process of Insight Control licensing to be very simple and straightforward. First, Insight Control is licensed per physical HP server. That means for every physical HP server you want monitored and managed by Insight Control, one software license is required.

Ah, but what if you use virtual servers? Then you're in for a bargain! Regardless of the number of virtualized hosts on a physical server, just one HP Insight Control license is required.

But remember that the Insight Control license is tied to that particular HP server and isn't transferrable to another HP server, unless that server is replaced due to a warranty/service event .

In order to use HP Insight Control on HP servers, you must purchase, redeem, and then deploy your HP Insight Control software license.



One server = one software license!

Purchasing a software license

HP Insight Control is available for purchase directly from HP and from HP channel partners. The best way to obtain an Insight Control software license is to purchase the software license at the same time you purchase your HP Server, but you can always purchase Insight Control software separately.

HP Insight Control software licenses are available in the following packaging options:

- ✓ **Single-License Packs:** Provides you with one Insight Control software license key per package.
- ✓ **Flexible Quantity Packs:** You select the number of software licenses needed and HP ships one License Entitlement to you that provides all the licenses you ordered.
- ✓ **Bundled with HP Servers:** You can buy Insight Control licenses bundled with select BladeSystem Enclosures and select ProLiant rack and tower server models.
- ✓ **Pay-As-You-Go Volume Licensing:** This simple volume licensing process (based on HP's Activation Key Agreement) provides a single Master License key for all your HP servers. The single master key eliminates the need for you to manage a unique software license key for each of your HP servers. (This program requires minimum purchase of 100 licenses.)

When you purchase HP Insight Control software, you receive a *License Entitlement* from HP. The License Entitlement can be in the form of an e-mail (electronic delivery) or an entitlement certificate (physical delivery). This License Entitlement must then be redeemed before you can begin using Insight Control.

Redeeming a license entitlement

After you purchase HP Insight Control, you must register that purchase on *Poetic* (HP's software license delivery system). By registering your software purchase with HP, you create software licenses and objects needed to deploy HP Insight Control, such as software license keys. The registration also triggers the creation of contracts for Insight Control software support bundled with your purchased software.



You must first redeem your license entitlement in order to get a software license key.

Deploying a software license

Once you receive your software license key, you can deploy your software. HP Insight Control software is deployed through the Managed Set-up Wizard or License Manager in HP Systems Insight Manager — resulting in a licensed server ready for use.

The Managed System Set-up Wizard (MSSW) feature provides a guided process to set up managed servers with Insight Control. The License Manager enables you to view and manage Insight Control software licenses within HP Systems Insight Manager (HP SIM). The License Manager also allows you to view and associate licenses to specific servers discovered by HP SIM.

This book doesn't go through all the installation details; for that you'll need the HP Insight Control user guide. You can also use installation services from HP and HP channel partners.



For more information on licensing, please visit http://h18013.www1.hp.com/products/servers/proliant_essentials/license/licenseops.html.

Software Support and Updates

Purchase of Insight Control includes one year of 24x7 HP Software Technical Support and Update Service. This service provides access to HP technical resources for assistance in resolving software implementation or operations problems. The service also provides access to software updates and reference manuals in electronic format as they're made available from HP. With this service, Insight Control customers will gain expedited problem resolution as well as proactive notification and delivery of software updates. For more information about this service, see www.hp.com/services/insight.

Once you register your HP Insight Control purchase, you receive a service contract in the mail containing the Customer Service phone number and the Service Agreement Identifier (SAID). You need your SAID when you call HP for technical support. Tech support uses your SAID to view your support contract online.

Besides access to HP Support, your active contract and SAID entitles you to get new versions of HP Insight Control software when releases come out. Quite often those new versions will contain significant enhancements and new features you'll find valuable to use on your existing servers. HP recommends you get your Insight Control software updates electronically, which you can do by linking your support contract SAID to your profile on the HP Support Center website (www.hp.com/go/hpsc).

HP Media Kits

Insight Control software is available on a DVD media kit. Media kits contain Insight Control software and documentation, but they don't include software licenses or activation keys. HP media kits aren't required but can be valuable if downloading software from hp.com isn't possible or cost-effective, or if you can't burn your own dual layer DVDs after downloading an image.



And it never hurts to have a backup, just in case.

Chapter 6

Ten Benefits of HP Insight Control

In This Chapter

- ▶ Speed your deployments and migrations
 - ▶ Improve your administrators' productivity
 - ▶ Reduce downtime
 - ▶ Increase your total capacity
-

If the decision-makers in your company need justification for using HP Insight Control and HP ProLiant servers, then this chapter is for you — and for them. A simplified version of this information is repeated from earlier chapters to help you make your case more effectively.

Faster Server Deployments

Testing by HP's engineering team as well as observations of ProLiant customers running Insight Control demonstrates a 12-fold increase in server deployment speeds using Insight Control compared to manually deploying a new server image. All you have to do is configure the server deployment once and then deploy as often as needed. The average deployment time drops from 4 hours to 20 minutes.

Faster Server Migrations

Migrating a server image to a new ProLiant server from your existing HP or non-HP server is now wizard-driven — you don't have to engage in manual steps, eliminating migration

errors. Insight Control lets you transfer the operating system, applications, and data without any errors, and also inject new drivers for the ProLiant server. Migrations can be three times faster than migrating to a server from the ground up.

Reduce Travel Expenses

Insight Control with iLO Advanced software provides server administrators with advanced server health and control information, and helps server administrators access and troubleshoot HP servers anywhere in the world, no matter their status or operating system state. The result? Reduced or eliminated travel costs, easily saving \$500 to \$1,500 per incident. Insight Control can pay for itself on the first use!

Reduce Unplanned Downtime

HP Systems Insight Manager with iLO can reduce downtime by as much as 83 percent. Insight Control alerts you to potential downtime issues quickly so you can resolve them as soon as possible. HP Systems Insight Manager integrates HP Smart Update for quick version control and software update distribution, significantly reducing downtime.

Triple Data Center Server Capacity

If power and cooling issues restrict your server capacity (especially during high energy use periods), then you need control and optimization of the electrical power used by your servers. Dynamic power capping in Insight Control limits power to over-provisioned servers and optimizes your data center capacity — increasing capacity by as much as three-fold, reducing your energy bills, and improving the uptime of your critical services.

Manage HP Servers from a Single Pane of Glass

Insight Control integrates seamlessly with Microsoft System Center, VMware vCenter Server, and HP IT Performance Suite (formerly Ops Center) software. You don't have to switch between different windows because Insight Control integrations allow you to manage your virtual and physical servers together from your management console of choice.

Test Insight Control for Free

Download and test a fully-functional version of HP Insight Control on your ProLiant servers for free. Go to the HP Insight Control website at www.hp.com/go/insightcontrol and click on the 'Free Trial Licenses' link to download a 60-day trial license of Insight Control. Install it, test it, and understand how Insight Control can save you money.

Technical Support and Software Updates

Every HP Insight Control license includes one year of unlimited technical support and access to all software updates. Further customer training, installation services, and assessment services are available. You can also view free white papers, educational videos, and other information at the HP Insight Control website at www.hp.com/go/insightcontrol.

Use Agentless Management

HP Insight Control 7 supports agentless management operation on ProLiant Gen8 servers. While HP Gen8 servers can be operational after connecting the power cable and a network cable, Insight Control adds capabilities for health and alerting and integrates HP Smart Update for maintaining version control across the many servers in your environment.

Enjoy Fast Payback, Higher ROI

An IDC white paper, published in March 2012, found that the efficiencies provided by Insight Control resulted in a reduction in costs of running ProLiant servers by \$73,482 per 100 users over the three-year period. The study found the payback period for using Insight Control was only 4.2 months and the ROI was an enormous 642 percent! Extrapolating just the cost savings into an IT organization responsible for 1,000 users would net a savings of some \$730,000, money an innovative and growing company could certainly use for higher budget priorities.



Click on the 'ROI Calculator' link at the HP Insight Control website (www.hp.com/go/insightcontrol) to take an online questionnaire and to evaluate the ROI that Insight Control can provide to your business.

Appendix

Resources

If you would like to learn even more about HP Insight Control and other HP Insight Management solutions, here are some great places to visit:

- ✓ www.hp.com/go/insightcontrol: The one-stop shop for HP Insight Control Information, such as brochures, white papers, and more
- ✓ www.hp.com/go/tryinsightcontrol: HP Insight Control 60-day trial
- ✓ www.hp.com/go/insight: Your portal to the HP Insight Management portfolio
- ✓ www.hp.com/go/sim: HP Systems Insight Manager
- ✓ www.hp.com/go/ilo: HP iLO Management Engine
- ✓ www.hp.com/go/insightonline/info: HP Insight Online
- ✓ www.hp.com/go/matrixoe: HP Matrix Operating Environment
- ✓ www.hp.com/go/insightswdemos: A variety of demos on HP Insight Management capabilities

Reduce costs and the time it takes to manage your server infrastructure

With HP Insight Control, you can simplify and automate many server management tasks, improve the efficiency of your server administrators, reduce the time it takes to deploy new servers, dramatically decrease unplanned server downtime and energy costs, and eliminate performance downtime. Insight Control maximizes the ROI of HP servers.

- **Understand server management** — *before learning the basics of HP Insight Control, first get the basics*
- **Discover the HP Insight Management portfolio** — *see how Insight Control helps deploy, monitor, control, and optimize server management*
- **Find out how easy it can be** — *Insight Control makes server administrators more efficient by simplifying and automating day-to-day server administration*
- **Get the lowdown on HP Insight Control benefits** — *all the details you need to know to understand what Insight Control can do for you*



Open the book and find:

- How HP Insight Control will benefit your organization
- A section on how Insight Control really works
- How Insight Control can help you manage HP servers and save time and money
- Information on how Insight Control is licensed and packaged
- Answers to your questions on HP Insight Control

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